

# Development and validation of a BCI-enabled Boccia ramp for sport participation

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**Abstract**—We present a brain-computer interface (BCI) system designed to enable individuals with severe motor disabilities to play Boccia, a Paralympic sport. Boccia is a precision sport in which the objective is to get a ball as close as possible to a target. In its most adapted form, Boccia allows for the use of a ramp to assist the user. The proposed system consists of a BCI-enabled ramp that can be controlled by the user's brain signals using a visual control paradigm (i.e., P300, or SSVEP). We developed a software interface using custom tools in Unity and Python for the front-end and back-end, respectively. To validate the software, we tested the system with five subjects who performed six pipelines (three with P300 and three with SSVEP) to simulate real-world use. Each pipeline consisted of 10 guided selections in the software. The classifiers used Riemannian geometry and shrinkage linear discriminant analysis (sLDA) for P300 and canonical correlation analysis (CCA) for SSVEP. The results showed that the P300 ( $93 \pm 3\%$ , mean  $\pm$  SEM) paradigm had higher classification accuracy than the SSVEP ( $27 \pm 0.02\%$ , mean  $\pm$  SEM) paradigm. Additionally, we designed and built a 3D CAD model and a hardware prototype of the ramp. The hardware prototype uses linear actuators to change the incline of the ramp and the height of the ball. Stepper motors allow for the rotation of the ramp and the release mechanism of the ball. Recommendations on improvements to the hardware and software components are made for future prototypes. The presented system opens new possibilities for sports applications that can improve the quality of life of people with severe motor disabilities.

**Index Terms**—EEG, Brain-Computer Interface, P300, SSVEP, Adaptive Sports, Pediatrics

## I. INTRODUCTION

Brain-computer interface (BCI) is an emerging neurotechnology that allows individuals to control external devices (e.g., computers, wheelchairs) with their electrical brain activity. Most BCI applications are researched with adults, specifically after a traumatic event or for a neurodegenerative disease (e.g., Amyotrophic lateral sclerosis, or multiple sclerosis). This could be because it is likely that the adult is cognitively intact and can easily follow instructions. Children, are a

more challenging population; their brains develop rapidly, and their disability might not allow them to communicate easily, or follow instructions [1]. However, it has been shown that children with stroke and quadriplegic cerebral palsy are able to use BCI applications [2]–[4]. Nevertheless, the challenges to work with such population remain.

Cerebral palsy (CP) is the most common cause of motor disability in childhood [5]. Overall CP prevalence was 2.5 per 1000 live births among children born from 2002 to 2017 in Ontario, Canada [6]. While incidence is expected to remain stable for a few years, the number of people with the condition is expected to increase to more than 94,000 individuals in Canada by 2031 [7]. This projected increase in individuals with CP calls for action to develop engineering applications, such as BCI, that can improve the quality of life of such individuals.

Individuals with severe physical disabilities, like severe quadriplegic CP (qCP), often have high cognitive capacity but are trapped inside their bodies, thus, deprived of their fundamental human right and opportunity to meaningfully interact with their world – including a child's right to play [8]. Participation in sports fosters a sense of belonging and inclusion and facilitates the development of peer relationships. However, individuals with severe physical disabilities are unable to participate in traditional sports. This exclusion from sports and associated social activities has shown negative impact in the child's development [9].

Paralympic sports allow individuals with a specific disability to participate in a modified version of a traditional sport. However, the majority of Paralympic sports still require the athlete to be able to perform a certain movement or communicate. This may be impossible for individuals with severe motor disability (e.g., qCP, stroke) and complex communication needs. Nevertheless, these individuals might be cognitively intact, yet unable to participate in Paralympic sports. To the best of our knowledge, there are no sports-related BCI applications in research or clinical scenarios.

Boccia is a Paralympic sport that could allow individuals with severe motor disability to play with others using a BCI

This work was funded by the HBI Brain CREATE and ACHRI Haslam postdoctoral fellowships, and by Alberta Innovates.

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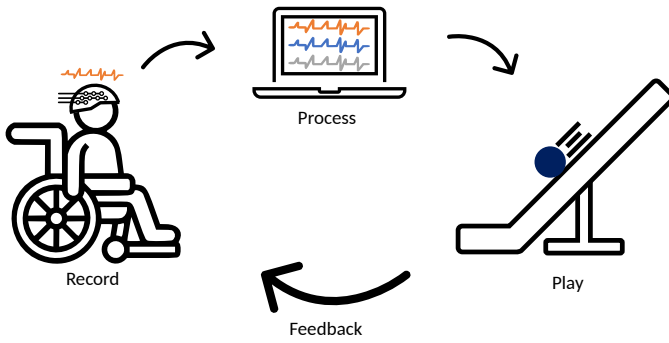


Fig. 1. Schematic of the proposed BCI system. The system is composed of the recording hardware, a computer for data digitization and processing, the mechanical ramp with actuators to enable independent play, and visual feedback to the user.

system. The goal of Boccia is to aim and launch a ball as close as possible to a target. Boccia offers several levels of accommodation for players with disabilities, primarily via an assistive ramp for aiming and launching the playing ball. Presently, in its most adaptive form, the player verbally instructs an assistant to move the ramp during the aiming phase. After the ramp is positioned, the player manually pushes or uses an adaptive mechanism (e.g., mechanical switch) to drop the ball down the ramp and towards the target. While this adaptation makes it possible for players with motor disability to play, it may not be sufficient for individuals living with the most severe disabilities (i.e., those who are unable to reliably communicate, have non-verbal communication only, or operate basic assistive devices due to physical limitations). Additionally, the need for an assistant reduces the players' independence during the game.

To address these limitations, this work aims to develop a BCI-enabled Boccia ramp to allow individuals with severe motor disabilities to play a sport independently. The objectives of this work are a) to present the overall concept of a BCI-enabled Boccia ramp and b) validate the software used to interact with said ramp through two traditional BCI visual control paradigms: the event-related potential P300, and steady-state visual evoked potential (SSVEP) control schemes. The P300 paradigm is related to the attention of the participant; subsequent stimulus are presented with the target being activated infrequently [10]. The SSVEP paradigm is a response to an external visual stimulus, typically elicited by using concurrent stimuli of different frequencies [11]. The work presented here is divided in two main sections, the software development and validation, and the hardware and firmware development.

## II. METHODOLOGY

### A. BCI System

The complete BCI system is composed of the software visual interaction with the participant, the signal processing toolbox, the firmware controller, and the ramp hardware (Fig. 1).

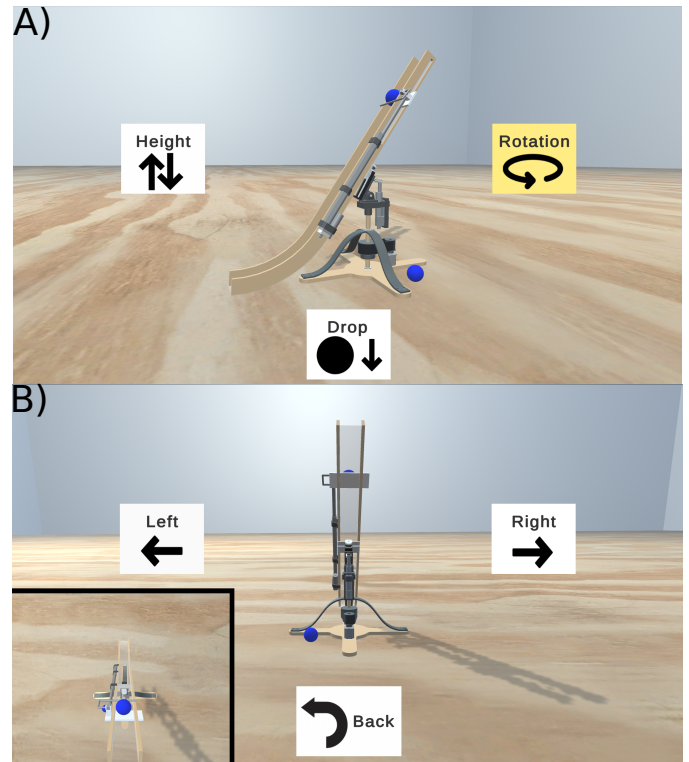


Fig. 2. Screenshots of the Unity-based software interface. A) Main screen showing the main 3 buttons: height, rotation, and drop. Rotation is highlighted in yellow to indicate to the user that they should focus on this button next. B) Secondary screen displayed after the rotation button has been selected. Notice that the buttons change to: left, right, and back, respectively. There is a birds-eye view on the bottom left corner to help the user understand the positioning of the ramp.

### B. Software

The software was developed using Unity (Unity Software Inc., San Francisco, USA). The BCI-essentials tool was used to enable the software to be controlled with BCI inputs [12]. The BCI-essentials tool allows objects in the Unity software to be used as stimulus presenting objects (SPOs). SPOs can change stimulation paradigm between P300 and SSVEP with a few selections in Unity's graphical user interface (GUI). This allows for rapid prototyping and comparison between the two paradigms. Both stimulus paradigms used 3 SPOs per screen with the main screen showing SPOs for height, rotation, and drop (Fig. 2A). The secondary screen would change according to whether height or rotation were selected (Fig. 2B). If height was selected, the secondary screen would change the buttons to up, down, and back; if rotation was selected, the secondary screen would change the buttons to left, right, and back. SPOs would toggle between a white (off) and a red (on) background color whenever the stimulus was active. The settings for the SPOs were the following:

- P300 had an on and off times of 50 and 100 msec, respectively. There were between 9 and 12 flashes for each selection.
- SSVEP response data was collected for 5-second windows before trying to do a classification. Stimulation

TABLE I

MOTORS USED AND DRIVERS IN THE BCI-ENABLED BOCCIA RAMP. THE TABLE INCLUDES MAX LOADS OR RADIAL FORCES, RESPECTIVELY, AND MAX SPEED. THESE METRICS WERE CHOSEN TO BE THE MOST IMPORTANT AS THEY ARE DIRECTLY RELATED TO THE HARDWARE CONSTRAINTS BASED ON THE CURRENT BOCCIA RULES.

| Motor model     | Driver           | Motor manufacturer      | Action        | Max load (l) or radial force (rf) | Max speed   |
|-----------------|------------------|-------------------------|---------------|-----------------------------------|-------------|
| FA-PO-150-12-2  | PChero L298N     | Firgelli automations    | Ramp incline  | 150 lbs (l)                       | 0.5 in/sec  |
| PA-14P-18-35    | PChero L298N     | Progressive automations | Ball height   | 35 lbs (l)                        | 1.38 in/sec |
| SCA5618L2804-A  | COVVY TB6600     | Nanotec                 | Ramp rotation | 75 N (rf)                         | 100 rpm     |
| XY42STH34-0354A | Longrunner A4988 | Adafruit Industries LCC | Ball release  | 25 N (rf)                         | 100 rpm     |

frequencies were 8.33, 10, and 12.5 Hz.

BCI-essentials-Python was used for the signal processing [13]. This tool integrates with its Unity counterpart to provide a complete BCI software system. The Python tool integrates with LabStreamingLayer (LSL) [14] to enable EEG data and Unity markers integration. Pre-processing included 4th order Butterworth bandpass filtering of the raw EEG traces. Cut-off frequencies ( $f_c$ ) were 0.1 and 10 Hz for P300, and 1 and 40 Hz for SSVEP. Online classification of the P300 trials was done by calculating the covariance matrices using a XDown spatial filter, taking the tangent space representation of the covariance matrices, and classifying them with shrinkage linear discriminant analysis (sLDA) [15]. Online classification of the SSVEP trials was done using a canonical correlation analysis (CCA) approach using the fundamental frequencies and their first 2 harmonics [16]. Additionally, the P300 trials included a calibration step at the beginning of the experiment. Calibration consisted of 12 trials with the same settings previously described.

### C. Experimental validation

To validate the software, data was collected from 5 subjects (1 male, 4 female)  $25.6 \pm 5.5$  years old [Mean  $\pm$  SD]. Data was acquired using a DSI24-C system sampling at 300 Hz (Wearable Sensing, San Diego, USA). Ethical approval was obtained from the University of Calgary's Conjoint Health Research Ethics Board (REB15-2567). Each subject did one session that consisted of three pipelines using P300 and SSVEP (i.e., 6 trials per session). The pipelines consisted of a set of 10 instructions guiding the subject through a typical positioning of the ramp. For example: select elevation, do 3 steps up, go back, select rotation, do 2 steps left, go back, drop the ball. The pipelines and the order of paradigm (i.e., P300 or SSVEP) were randomized to avoid any bias. Subjects were allowed a minimum of 2 minutes between pipelines to rest their eyes. If the BCI system or the subject caused an incorrect selection of the next step in the pipeline, the software would add additional selections for fix said mistake. For example, if the subject had to select "rotation" and the "elevation" was selected instead, the system would automatically update and guide the subject to select "back" and then "rotation" to fix such mistake.

To quantify the performance of the subjects using both paradigms, we calculated the number of correct and incorrect selections in each trial. Additionally, we computed the total

accuracy per trial considering all the correct and incorrect selections made. This metric allow us to compare how good would the subject perform using the different paradigms as if they were using the ramp in a real-world scenario (i.e., where one could make some mistakes in the positioning of the ramp, and would have to make up for them).

The monitor used was a 27" LG27GL850 (LG Corporation, Seoul, South Korea) configured with 100 Hz refresh rate. This refresh rate was selected so that the stimulation frequencies could be generated by using full frames (e.g., 8.33 Hz is achieved by switching the image every 12 frames). The stability of the chosen SSVEP frequencies was verified using a custom photo-sensor placed on the screen to validate that the expected frequency was actually produced by the Unity software.

### D. Hardware & firmware

The hardware of the ramp was modeled after commercially available ramps like the Boss 360 from Gravity Boccia (Therafin Corporation, Frankfort, USA). First, a computer-aided design (CAD) 3D model of all the ramp components was created using Fusion 360 (Autodesk, San Francisco, USA). All the components were modelled to ensure the dimensions of the ramp would comply with the current international Boccia regulations. Additionally, the range of motion of all motors was simulated to ensure the motors would satisfy their movement requirements. The rails and base were build of 5/8" treated plywood, and the main shaft was built with 1" diameter oak wooden dowel to ensure that the structure would be light and stable. The back of the rails was built with 1/8" clear acrylic to allow the players to see the ball roll. Lastly, the main supports were built with 1/8" steel to provide the require stability. Lastly, custom holders for the motors were designed and 3D printed using a PRUSA i3 MK3s (PRUSA Research, Prague, Czech Republic) with 1.75 mm PLA plastic. The previously mentioned characteristics were recommended to us by players and coaches from the Boccia Canada team, players from the local Boccia league, and organizer of the local Boccia league from the Mount Royal University and the Calgary Adapted Hub.

The motors and drivers selected to control the movements of the ramp are summarized in Table I. The incline actuator was chosen based on the expected weight of the ramp and comments from one local athlete that recommended the ramp only inclines about 5 cm (i.e., 2 in) since most of the time, the

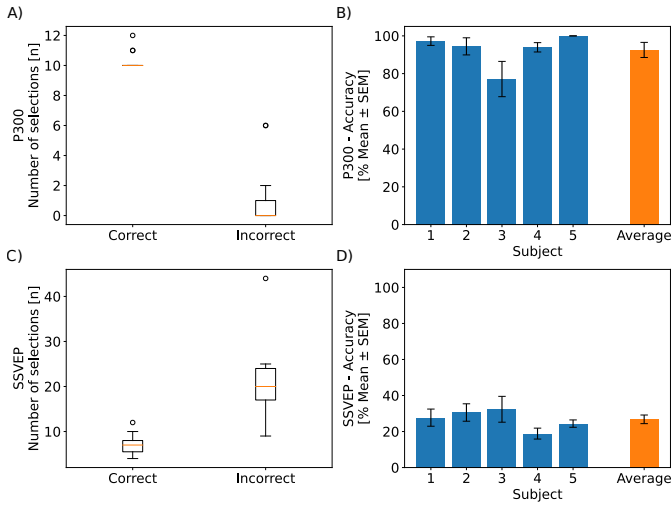


Fig. 3. Left panels show the total number of correct and incorrect selections using the P300 (A) and SSVEP (C) paradigms. Right panels show the accuracy for the total number of selections (i.e., including added steps to correct deviations from the initial pipeline) using the P300 (B) and SSVEP (D) paradigms.

edge of the ramp is touching the floor. The ball height actuator was chosen due to its 45 cm (i.e., 18 in) stroke length. The stepper motor used to rotate the ramp was chosen due to its 75 N radial force, and its 2.8 A winding current at 24 V. Lastly, the stepper motor used in the ball release mechanism was chosen due to its 25 N radial force while maintaining a small (NEMA 17) chassis, and low weight (220 gr).

The firmware of the ramp was implemented using an Arduino Mega 2650 Rev 3 (Arduino, Sommerville, USA). This Arduino model is based on the ATmega2560 microcontroller unit (MCU). This model was selected due to the extra interrupt-enabled pins (i.e., 6 pins in total, compared to the 2 pins available in the Arduino Uno board). The firmware was written in C/C++ using the PlatformIO extension in VS Code (<https://platformio.org/>). An Object-oriented programming approach was preferred to create objects for each type of motor (i.e., stepper motor or linear actuator). The motors can be controlled by sending a command over serial interface using the USB connection from the computer to the Arduino. The serial command is sent to the Arduino through USB from the software previously described in section II-A.

All motors perform a calibration step after the power supply has been turned on; calibration is made for each motor subsequently. The stepper motors are associated with optical switches placed in the expected motion range, before any mechanical failure point (e.g., the release mechanism switches are placed in a range before the lever would hit the wooden rails). The optical switches are connected to the Arduino's interrupt pins. The stepper motors rotate counter-clockwise until one of the optical switches is activated. Then, the stepper motors rotate clockwise until the second optical switch is activated. The step positions where the switches are activated are stored by the Arduino to determine the

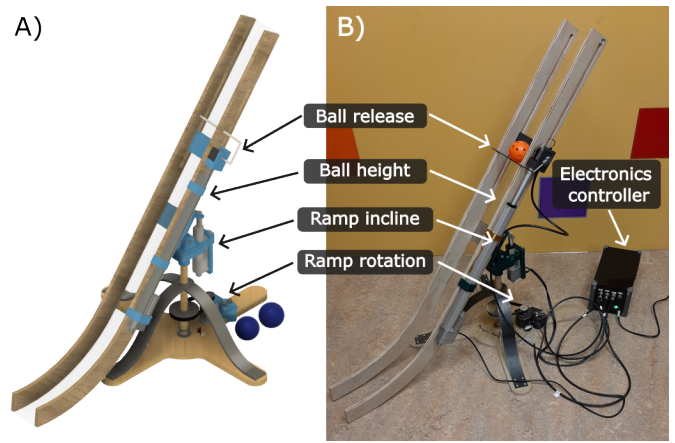


Fig. 4. A) 3D CAD model of the Boccia ramp. B) Hardware prototype of the Boccia ramp along with electronics controller. Labels included highlight the motors used for ball release, ball height, ramp incline, and ramp rotation.

operational movement range. If the software sends a command to move the stepper motors over the predetermined ranges, the command is changed to avoid going over the range. If the switch gets activated due to an external factor (e.g., the ramp being hit by another player), the motion range is recalculated based on the current step position of the stepper motor. The calibration of the actuators is done by measuring the resistance of the potentiometer inside the actuator. The actuators are set to extend and retract until the value of the potentiometer is stable. This provides the potentiometer values at full range of motion.

### III. RESULTS

#### A. Experimental validation

The results for the number of selections with the P300 paradigm (Fig. 3A) shows a minimum of 10 selections because all the subjects were able to finish the pipelines with said paradigm. Meanwhile, the SSVEP paradigm (Fig. 3C) shows a lower minimum because some subjects asked for the experiment to finish earlier due to the frustration of having too many incorrect selections.

The overall accuracy were  $93 \pm 3 \%$  and  $27 \pm 0.02$  [Mean  $\pm$  SEM] for the P300 and SSVEP paradigms, respectively. Note that these accuracy metric includes the added steps to correct deviations from the original pipelines. The accuracy with the P300 paradigm (Fig. 3B) shows higher values than SSVEP (Fig. 3D). The increase in accuracy can be seen for each individual subject, as well as the average.

#### B. Hardware & firmware

The 3D CAD model can simulate movement in the desired directions and the individual parts can be exported as engineering drawings for cutting of the parts and constructing the ramp (Fig. 4A). The prototype of the Boccia ramp was built following the 3D CAD model (Fig. 4B).

## IV. DISCUSSION

### A. Experimental validation

The experimental validation showed that the subjects were able to use both the P300 and SSVEP paradigms with the Boccia software. The increased number of incorrect selections (Fig. 3C), and lower accuracy (Fig. 3D) using the SSVEP paradigm could be attributed to the positioning and size of the elements on the screen. Previous reports have shown that closer stimulus can be the sensitivity of different parts of the visual field may depend on temporal frequency [11]; this means that temporal-frequency differences are mixed with spatial position and can cause inaccurate classification in the BCI system. Further research is needed to determine the optimal positioning and visual stimulus to enable accurate selections without causing unnecessary discomfort to the user.

Anecdotal, participants reported getting frustrated whenever their selection was incorrect, especially while using the SSVEP paradigm. This frustration could distract the participant from the task at hand (i.e., controlling the ramp); adding to the distractions in a real game (e.g., other people making noise, bright lights). Previous studies have shown that distractions can decrease the accuracy in SSVEP classification [17]. The increase frustration could have impacted the SSVEP classification. However, this is purely speculative and should be further explored.

Additionally, the classifier used for the SSVEP was a CCA-based classifier [16]. While CCA classification tends to provide good results future experiments could test other types of classifiers. For example minimum energy combination (MEC) [18] and multivariate synchronization index (MSI) [1] have proven to give high classification rates in online applications using as short as 1-second windows.

The study presented initial validation of the BCI software. However, the ramp is intended to be used by people with different types of disabilities. Future studies should investigate its usability with these populations, including pediatric users.

### B. Hardware & firmware

The hardware prototype was developed according to the 3D CAD design (Fig. 4). The hardware consists of the mechanical components, as well as the actuators and electronics controller. The ramp allows for the most common ranges of motion used in Boccia (i.e., ramp incline, and rotation, and ball height). The ball release mechanism does break the current Boccia regulations that state that the athlete must be in physical contact with the ball at the moment of release. However, as this device is not intended for professional use for now, this limitation is inconsequential.

More significant limitations of the hardware include the total height of the ramp (1.15 m); for comparison, the Boss 360 Gravity Boccia ramp with its 60.96 cm (24") adapter provides a total height up to 1.91 m. The height of our ramp was constrained by the elevator actuator to 45.72 cm (18") stroke. Future prototypes could incorporate a telescopic mechanism that allows the ramp to further extend to match the height

of the commercial ramps. The actuators require substantial voltage (24 V) and current (up to 4 A), requiring a benchtop power supply. If a different mechanical design leverages the use of gears that would reduce the torque and power requirements, smaller actuators and power supply could be used to allow for a battery-operated design. This battery-operated design could allow for easier use of the ramp in scenarios where a power outlet is not easily available.

## V. CONCLUSIONS

In this study, we developed a BCI-enabled Boccia ramp system and validated the software controller with 5 healthy adult subjects. We compared P300 and SSVEP paradigms and found that the P300 paradigm showed higher accuracy in the experimental validation. This proposed system is, to the best of our knowledge, the first implementation of a BCI application aimed towards a sports application. Further development and testing are required to improve the ramp prototype and determine the right parameters of the software interface (e.g., user experience, user interface, other types of paradigms for control). Nevertheless, the current implementation of the system opens up possibilities to investigate new BCI applications that allow individuals with severe motor disabilities to participate in their communities in an independent manner.

## REFERENCES

- [1] J. Zhang, Z. Jadavji, E. Zewdie, and A. Kirton, "Evaluating If Children Can Use Simple Brain Computer Interfaces," *Frontiers in Human Neuroscience*, vol. 13, Feb. 2019.
- [2] E. D. Floreani, D. Rowley, N. Khan, D. Kelly, I. Robu, A. Kirton, and E. Kinney-Lang, "Unlocking Independence: Exploring Movement with Brain-Computer Interface for Children with Severe Physical Disabilities," *Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Annual International Conference*, vol. 2021, pp. 5864–5867, Nov. 2021.
- [3] E. D. Floreani, D. Rowley, D. Kelly, E. Kinney-Lang, and A. Kirton, "On the feasibility of simple brain-computer interface systems for enabling children with severe physical disabilities to explore independent movement," *Frontiers in Human Neuroscience*, vol. 16, 2022.
- [4] Z. Jadavji, J. Zhang, B. Paffrath, E. Zewdie, and A. Kirton, "Can Children With Perinatal Stroke Use a Simple Brain Computer Interface?," *Stroke*, vol. 52, pp. 2363–2370, July 2021. Publisher: American Heart Association.
- [5] J. R. Wolpaw, N. Birbaumer, W. J. Heetderks, D. J. McFarland, P. H. Peckham, G. Schalk, E. Donchin, L. A. Quatrano, C. J. Robinson, and T. M. Vaughan, "Brain-computer interface technology: a review of the first international meeting," *IEEE transactions on rehabilitation engineering: a publication of the IEEE Engineering in Medicine and Biology Society*, vol. 8, pp. 164–173, June 2000.
- [6] A. Ahmed, L. C. Rosella, M. Oskoui, T. Watson, and S. Yang, "Prevalence and temporal trends of cerebral palsy in children born from 2002 to 2017 in Ontario, Canada: Population-based cohort study," *Developmental Medicine & Child Neurology*, vol. 65, no. 2, pp. 243–253, 2023. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/dmcn.15324>.
- [7] N. Amankwah, M. Oskoui, R. Garner, C. Bancej, D. G. Manuel, R. Wall, P. Finès, J. Bernier, K. Tu, and K. Reimer, "Original quantitative research - Cerebral palsy in Canada, 2011–2031: results of a microsimulation modelling study of epidemiological and cost impacts," *Health Promotion and Chronic Disease Prevention in Canada : Research, Policy and Practice*, vol. 40, pp. 25–37, Feb. 2020.
- [8] U. G. Assembly, "Convention on the Rights of the Child," *United Nations, Treaty Series*, vol. 1577, no. 3, pp. 1–23, 1989.

- [9] N. Shields and A. Synnot, "Perceived barriers and facilitators to participation in physical activity for children with disability: a qualitative study," *BMC pediatrics*, vol. 16, no. 1, pp. 1–10, 2016. Publisher: BioMed Central.
- [10] S. J. Luck, *An introduction to the event-related potential technique*. MIT press, 2014.
- [11] A. M. Norcia, L. G. Appelbaum, J. M. Ales, B. R. Cottareau, and B. Rossion, "The steady-state visual evoked potential in vision research: A review," *Journal of Vision*, vol. 15, p. 4, May 2015.
- [12] B. Irvine, E. Kinney-Lang, M. Wehner, and G. Wilding, "BCI-essentials-unity," <https://github.com/kirtonBCIlab/bci-essentials>, Mar. 2023.
- [13] B. Irvine, E. Kinney-Lang, A. Gulamhusein, G. Wilding, and D. Comaduran Marquez, "BCI-essentials-python," <https://github.com/kirtonBCIlab/bci-essentials-python>, Mar. 2023.
- [14] T. Stenner, C. Boulay, M. Grivich, D. Medine, C. Kothe, tobiashertzke, chausner, G. Grimm, xloem, A. Biancarelli, B. Mansencal, P. Maanen, J. Frey, kyucrane, S. Powell, P. Clisson, and phfix, "Labstreaming layer," <https://zenodo.org/record/7637592>, Feb. 2023.
- [15] F. Lotte, L. Bougrain, A. Cichocki, M. Clerc, M. Congedo, A. Rakotomamonjy, and F. Yger, "A review of classification algorithms for EEG-based brain-computer interfaces: a 10 year update," *Journal of Neural Engineering*, vol. 15, p. 031005, June 2018.
- [16] Z. Lin, C. Zhang, W. Wu, and X. Gao, "Frequency recognition based on canonical correlation analysis for SSVEP-based BCIs," *IEEE Transactions on Biomedical Engineering*, vol. 54, pp. 1172–1176, June 2007. Conference Name: IEEE Transactions on Biomedical Engineering.
- [17] Y. Zhao, J. Tang, Y. Cao, X. Jiao, M. Xu, P. Zhou, D. Ming, and H. Qi, "Effects of Distracting Task with Different Mental Workload on Steady-State Visual Evoked Potential Based Brain Computer Interfaces—an Offline Study," *Frontiers in Neuroscience*, vol. 12, 2018.
- [18] O. Friman, I. Volosyak, and A. Graser, "Multiple Channel Detection of Steady-State Visual Evoked Potentials for Brain-Computer Interfaces," *IEEE Transactions on Biomedical Engineering*, vol. 54, pp. 742–750, Apr. 2007. Conference Name: IEEE Transactions on Biomedical Engineering.